Anthony Coffey

Senior Product Engineer · AI in Regulated Healthcare
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SUMMARY

Senior product engineer with 12+ years of experience shipping full-stack products, specializing in production-ready AI features: LLM agents with live tools, RAG over private data, and OCR pipelines. Recent work includes PHI-redacting OCR for an FDA-regulated ultrasound AI platform, a fine-tuned GPT-4o-mini scheduling assistant, and a Gemini-powered submission review system approved by PepsiCo's CISO. TypeScript and Python across React, Node, FastAPI, Postgres, AWS and GCP. I work spec-first and test-driven with coding agents like Claude Code.

CORE SKILLS

- **AI features:** LLM agents & tool use (MCP) · RAG (Qdrant, pgvector) · fine-tuning · structured outputs · Anthropic, OpenAI and Gemini APIs · LiteLLM · OCR/vision (EasyOCR, Cloud Vision)
- **Frontend:** React 19 · TypeScript · Next.js · Vite · Tailwind / shadcn/ui · visx / Highcharts · React Native / Expo · Capacitor · Three.js (R3F) · i18n
- **Backend:** Python (FastAPI) · Node/Express · PostgreSQL · Redis/BullMQ · Firebase · Stripe · Twilio
- **Quality & Delivery:** Vitest · Playwright · pytest · GitHub Actions · Docker · AWS · GCP · Cloudflare · Vercel
- **Domain:** HIPAA/SOC 2 product requirements · FDA 510(k) / SaMD · multi-tenant SaaS · localization (EN/ES/PT)

EXPERIENCE

Full Stack Software Engineer — Bitmotive, LLC · Contract

Client: Ultrasound.AI (AI delivery-date prediction from prenatal ultrasound) · Feb 2026 – Aug 2026

- Turned static matplotlib model-validation images into **interactive analytics** (visx zoom, brush and tooltips) for Bland-Altman, scatter and error-distribution charts. Added metric cards (MAE, RMSE, R^2, normality) and **one-click PDF reports** for clinical and regulatory teams.
- Built the backend of an **OCR pipeline that reads burned-in text** on ultrasound frames. It recovers estimated delivery dates the product was missing and **redacts patient identifiers** before images are stored.
- Shipped end-to-end **MFA** (setup wizard, challenge flow, recovery codes, institution-level enforcement) and a **searchable audit-log viewer** with CSV export for global admins.
- Localized the frontend (i18next) and FastAPI backend (Babel) in **English, Spanish and Portuguese**.
- Migrated E2E testing from Cypress to **Playwright (200+ tests),** configured to run against production build in CI.

Lead Software Engineer — 24hrcarunlocking.com
Aug 2023 – May 2026

- Shipped **RescueBot**, an in-app scheduling assistant on a **fine-tuned GPT-4o-mini** model that collects service details conversationally, quotes from live pricing, and hands off to a human on request.
- Built **AI-generated price quotes**: an LLM classifies the vehicle and normalizes the requested time from free text, then business rules price the job.
- Architected **AmeniLink**, a multi-tenant property-management SaaS for web, iOS and Android (React 19, TypeScript, MUI, Firebase, Capacitor) with invitations, RBAC and service requests.
- Built the dispatcher portal (Nuxt/Vuetify), technician mobile app (React Native/Expo) and Node/Express API, including **real-time auto-dispatch** for ~14,000 calls and ~15,000 SMS a month, push/SMS notifications, Stripe payments and lead intake webhooks.
- Rebuilt the marketing site on Astro + Sanity + Cloudflare (**~$6,700/yr saved**, 99+ PageSpeed). Consolidated 913 legacy pages into ~21 canonical pages and added Sanity AI agent actions for content.

Staff Engineer — Southern Made · Contract
Apr 2024 – Sep 2024

- Designed the architecture for the **DORITOS® Crash the Super Bowl** submission platform on GCP, with Gemini and Cloud Vision automatically reviewing user-generated entries. The design was **approved by PepsiCo's CISO (Chief Information Security Officer)**.

Full Stack Software Engineer — Marani Health
Mar 2022 – Apr 2024

- Built a React annotation app (AWS Amplify) where physicians labeled anonymized fetal and maternal heart-rate data to validate the algorithm behind an **FDA 510(k) submission**. Raised Python algorithm test coverage **from 0% to 99%**.

Cybersecurity Analyst — Health Tech Defenders · Contract
Dec 2023 – Sep 2025

- Penetration-tested healthcare web apps and APIs and delivered remediation guidance for HIPAA-regulated environments.

Earlier Experience · 2014 – 2022

- **React Native Developer, festivalPass** (2021–2022): Developed cross-platform React Native app at full parity with web, backed by Node/Strapi; owned store releases.
- **Freelance Engineer, Lifecycle Insights** (2019–2021): Vue/Node/MongoDB survey app with live data visualization and styled PDF reports.
- **Freelance Engineer, Smart Insurance** (2016–2019): React workers'-comp quoting app with multi-state underwriting rules fully compliant with state regulations.
- **Full Stack Developer, RiskPro®** (2014–2016): Developed Highcharts visualization layer and risk algorithms on a Python backend for a fintech platform on AWS/Lambda.

PROJECTS

- **Bugsy — Personal AI agent:** RAG over my repos and docs, live Jira, GitHub and Notion tools through MCP, an LLM-scored job board, and an issue-to-PR coding agent, all self-hosted on GCP.
- **coffey.codes:** Next.js 16 portfolio with a React Three Fiber 3D homepage, MDX blog (23 articles), Playwright/Vitest CI and synthetic analytics monitoring.
- **L3IQ (client project):** manufacturing label scanner. Cloud Vision OCR plus Gemini turns label photos into structured fields that pre-fill the client app, replacing manual data entry and streamlining quality control operations.
- **Simply Voice:** text-to-speech SaaS (React, Firebase, Google Cloud TTS) with Stripe subscriptions and usage quotas.
- **Flutter LabelScan:** mobile price-label scanner using Cloud Vision + Gemini to extract prices and keep a running cart total.
- **Periscope:** TypeScript CLI unifying Search Console, GA4, Bing and Google Ads keyword data.

EDUCATION

Gulf Coast State College — Associate of Arts · 2012